Dark-portal interactions of high-energy astrophysical neutrinos

Working notes

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1 Summary

What the code computes, and where it stands. We compute the diffuse flux of high-energy astrophysical neutrinos at IceCube in the dark-portal model of the theory notes (`notes/main.pdf`, "Notes on the theoretical framework for generalized new neutrino interactions"). A dark vector boson Z_D couples to active neutrinos through their mixing with a sterile state and to matter through Z–Z_D mass mixing. Two effects follow: neutrinos scatter resonantly on the relic (cosmic) neutrino background, which absorbs and regenerates them, and neutrons source a long-range coherent potential that changes their flavor in flight. The solver integrates both across cosmological distances, the Milky Way and the Earth; it is validated at four benchmark parameter points and runs in 0.32 s at its production defaults. The statistical analysis fits the predicted flavor composition to a projected 2050 measurement; it carries no explicit systematic uncertainties yet. The next stage, agreed on 2026-10-12, adds IceCube event rates, systematic uncertainties and coverage tests (Sec. 6). On 2026-10-12 we corrected the relic-neutrino temperature and every neutron number that sources the potential (Appendix A); a chord-sampling floor in the Earth transit remains open (Sec. 7).

Table 1 lists the state of each component. Every number in these notes comes from a run of the code, identified by the commit that contains it or by the test that checks it; anything not yet traced is marked **[VERIFY]**. Corrections to the code and to earlier statements are kept in Appendix A, because they are the reason the remaining numbers can be trusted. The agreed order of work is also kept, in shorter form, in `docs/dev/WORK_PLAN.md`.

Table 1: **State of each component, as of 2026-10-12.** "Validated" means checked at the four benchmark points of Sec. 4 with the acceptance rules stated there.

Component	State	Section
Flux solver	Validated; production defaults run in 0.32 s on 8 cores	3, 4
Thermal absorption rates	Semi-analytic kernel; relic temperature corrected 2026-10-12	3.2, A.1
Earth transit	Phase-exact stepping with a step-count certificate; 25 steps	3.5
Flavor-ratio analysis	Profile scans and exclusion planes run; no explicit systematics	5
IceCube event rates	Planned; data sets and folding method chosen	6.1
Systematics and coverage	Planned; list and procedure agreed	6.2, 6.3
Neutron densities	All corrected 2026-10-12	2.4, A
Earth-chord sampling	Floor of $\sim 10^{-2}$ in flavor fractions; open	7

2 The model and its implementation

2.1 Where the physics is defined

The theory notes define the framework; we cite their equations as "Eq. (Tn)". The neutrino ensemble is a 3×3 density matrix $\rho(E_0, z)$, where E_0 is the energy observed at Earth and z the redshift. It obeys the quantum kinetic equation (T3). The Hamiltonian (T4) drives its coherent part; the collision operator (T6) is its incoherent part. The coherent part contains the standard matter potential (T5) and the long-range dark-portal potential (T9) for neutrinos, (T10) for antineutrinos. The incoherent part contains the attenuation rate (T7) and the regeneration kernel (T8), evaluated with a Breit–Wigner resonance of width (T11) and coupling (T12). The

sources follow a power law with a free flavor composition (T18), distributed in redshift as the star-formation rate (T19, T20). The thermal (Debye) mass that screens the mediator is (T23).

Parameters.—The dark sector is set by the mediator mass m_{Z_D}, the dark gauge coupling g_D, the Z–Z_D mass mixing ϵ', and the active–sterile mixing magnitudes $u_\alpha \equiv |U_{\alpha 4}|$ ($\alpha = e, \mu, \tau$). Resonant absorption on a relic neutrino of mass m_i peaks near $E_{\rm res} = m_{Z_D}^2/(2m_i)$, so a mediator of mass $\sim$ 1–10 MeV places the absorption features inside the IceCube energy range. Table 2 lists every parameter with its default value and configuration key. The defaults live in one place, `src/dp_config.py` (`DEFAULT_CONFIG`); the analysis configurations carry their own copies of the physical constants.

Table 2: **Parameters and their defaults.** Keys refer to the solver's configuration sections; the analysis configurations use the same names under `physics_constants`. Masses and energies in eV unless stated.

Symbol	Meaning	Default	Key
m_{Z_D}	Dark mediator mass	2×10^6	`dark_portal.m_Zd_eV`
g_D	Dark gauge coupling	10^{-4}	`dark_portal.g_D`
ϵ'	Z–Z_D mass mixing	0.1	`dark_portal.epsilon_prime`
u_e, u_μ, u_τ	Active–sterile mixing	$0.5, 0.5, 0$	`dark_portal.u_*`
p	Source spectral index	2.5	`source.p_index`
$f_{e,S}, f_{\mu,S}, f_{\tau,S}$	Source flavor composition	$1/3, 2/3, 0$	`source.f_*_S`
$z_{\rm max}$	Largest source redshift	4	`source.z_max`
$m_{\rm lightest}$	Lightest neutrino mass	0.01	`cosmology.m_lightest_eV`
$T_{\nu,0}$	Relic-neutrino temperature today	1.676389×10^{-4}	`cosmology.T_nu0_eV`
$n_{n,0}$	Cosmological neutron density	3.091×10^{-8} cm^{-3}	`cosmology.n_n0_cm3`
$\theta_{12}, \theta_{13}, \theta_{23}$	Mixing angles	$33.41^\circ, 8.58^\circ, 49.1^\circ$	`standard_mixing.th*_deg`
$\delta_{\rm CP}$	Dirac phase	132°	`standard_mixing.d_cp_deg`
$\Delta m^2_{21}, \Delta m^2_{31}$	Mass splittings [eV^2]	$7.41 \times 10^{-5}, 2.511 \times 10^{-3}$	`standard_mixing.dm*_sq`

2.2 From the equations to the code

The solver (`src/`, entry point `run_solver` in `src/dark_portal_solver.py`) divides the propagation into three phases. Each phase hands its result to the next, and neutrinos and antineutrinos run separately before their fluxes are combined.

Cosmological phase.—From $z_{\rm max} = 4$ to $z = 0$ the flux is carried as populations in the instantaneous eigenbasis of the Hamiltonian (T4), one per mass state, with no coherences between them. At each step the populations decay at the thermal absorption rate (T7) and receive the regenerated flux of the cascade kernel (T8) and the injected source flux (T18). We project them onto flavor with the mixing probabilities $|U_{\alpha i}(E, z)|^2$ of the full Hamiltonian. The cascade kernel is separable, which reduces the regeneration integral to a suffix sum that costs $O(N_E)$ per step instead of $O(N_E^2)$ (`step_cosmological_propagation` in `src/dp_cosmo.py`).

Milky Way phase.—At $r_{\rm start} = 100$ kpc from the Sun the populations become a density matrix, which we evolve under the vacuum Hamiltonian plus the dark-portal potential sourced by the neutrons of the Galaxy, the Sun, the Moon and the Earth. We average the result over $n_{\rm MW}$ arrival directions (six by default), each integrated in $N_{\rm MW}$ steps (`src/dp_spatial.py`).

Earth phase.—The averaged density matrix is then evolved along $n_\oplus$ chords through the Earth (PREM densities, optional standard matter potential), from the surface to a detector 1.5 km deep. The chords are evenly spaced in the cosine of the zenith angle between straight up-going (-1) and horizontal (0); the flux returned is the plain average over them. The flavor fluxes at the detector are the diagonal of the averaged density matrix.

Antineutrinos.—The antineutrino run uses $U^*_{\rm PMNS}$ and the transformed potential $-V^*$ (T10). With the default `is_antineutrino = "sum"` the two runs are added, weighted by `nubar_to_nu_ratio` ($= 1$).

2.3 Potential textures

The flavor structure of both coherent potentials can be replaced through the configuration: `dark_portal.U_LRI_struct` for the dark-portal potential (default $u_\alpha u_\beta^*$, as in Eq. (T9)) and `standard_mixing.V_SM_struct` for the standard matter potential (default $\mathrm{diag}(1,0,0)$, as in Eq. (T5)). Each is a 3×3 nested list whose entries are real numbers or [Re, Im] pairs. The regression tests `test_structure_override_*` and `test_parse_structure_matrix_3x3_formats_and_validation` check the parsing and the antineutrino convention. In addition, we ran rank-zero, full-rank complex and indefinite textures through both spatial phases and the cosmological mixing table (2026-07-29); any fixed Hermitian texture works.

Limits of the override.—The texture must be Hermitian to 10^{-12} relative precision; the parser rejects anything else, because the propagator reads only one triangle of the Hamiltonian. It must be constant within a run, since the cosmological mixing table assumes a Hamiltonian of the form $M + \sigma S$ with one fixed matrix S. Overrides are given in the neutrino convention; the solver applies the antineutrino transformation itself. The override changes only the coherent potential: the Debye mass and the absorption rates remain set by u_e, u_μ, u_τ. Its overall scale is degenerate with $g_D\epsilon'$.

2.4 Physical inputs

Relic neutrinos.—The absorption rates integrate the Fermi–Dirac distribution of the relic neutrinos at temperature $T_{\nu,0}(1+z)$. Therefore, the relic density in our rates follows from $T_{\nu,0}$ and scales as $T_{\nu,0}^3$. The nominal 56 cm^{-3} per state enters only as a ratio that carries flavor asymmetries. The temperature is $T_{\nu,0} = (4/11)^{1/3}\,T_{\mathrm{CMB}} = 1.9454$ K $= 1.676389\times10^{-4}$ eV for $T_{\mathrm{CMB}} = 2.7255$ K. Until 2026-10-12 the code used 1.95×10^{-4} eV (Appendix A.1).

Neutron densities.—The dark-portal potential is proportional to the neutron number density along the path, because the Z-like couplings of protons and electrons cancel in neutral matter. Cosmologically, the neutrons sit almost entirely in helium-4, $Y_p/2$ per baryon, so $n_{n,0} = n_b Y_p/2 = 3.091\times10^{-8}$ cm^{-3} for $n_b = 2.515\times10^{-7}$ cm^{-3} (PDG 2024) and $Y_p = 0.2458$ (LBT Y_p project, 2026). In the Galaxy we convert the stellar mass density at 0.20 neutrons per nucleon, the electron density of the hot halo at 0.15 neutrons per electron, and the cold HI and H_2 discs of McMillan (2017) at 0.142 neutrons per nucleon (Appendices A.2 and A.4). The Sun, the Moon and the Earth also act as point sources, with 1.929×10^{56}, 2.232×10^{49} and 1.849×10^{51} neutrons. In the Earth we use $1-Y_e$ neutrons per nucleon, with electron fractions $Y_e = 0.4656$ in the core and 0.4957 in the mantle and crust, and the composition of water for PREM's top layer (Appendix A.3).

3 Numerical method

3.1 Resolution and defaults

Table 3 compares the solver's production defaults with the settings of the statistical analysis. The defaults follow the convergence study of 2026-07-28 (`tools/convergence_study.py`, results in `output/convergence/`). The analysis uses a coarser energy and redshift grid but more Milky Way paths and Earth chords, because its scans reach mediator masses beyond the study's parameter points.

The Earth-chord floor.—The Earth transit imprints interference fringes in energy that are finer than the energy grid. Therefore, at in-band parameter points, changing the number of chords moves the energy-averaged flavor ratios pseudo-randomly at the 10^{-2} level instead of converging. We treat this scatter as a numerical systematic of the unsmeared observable (`README.md`, Sec. 11). Folding through a detector's energy resolution, as the event rates of Sec. 6.1 will do, smooths these fringes.

Table 3: **Resolution settings.** The convergence column states what each setting is known to achieve.

Setting	Solver default	Analysis	Evidence
Energy bins N_E (10 TeV–30 PeV)	1000	200	$\sim 6 \times 10^{-3}$ on flavor ratios at 200
Redshift steps N_z ($z \leq 4$)	1000	200	$\sim 1.1 \times 10^{-2}$ at 200
Milky Way paths n_{MW}	6	50	
Earth chords $n_\oplus$	12	50	Fringe floor $\sim 1.1 \times 10^{-2}$ at 50 (in-band)
Milky Way steps N_{MW}	25	25	Converged to $\sim 10^{-10}$
Earth steps N_{Earth}	25	25	Certified, Sec. 3.5

3.2 Thermal absorption rates

The rates (T7) require a momentum integral over the relic distribution of a Breit–Wigner resonance whose width is far smaller than any practical grid. We integrate it semi-analytically: the momentum at which the resonance enters the angular range has a closed form, the narrow zone around it is integrated with exact antiderivatives, and the smooth remainder with Gauss–Legendre quadrature (`src/dp_thermal.py`). Against a dense reference integration, the significance-weighted error is 2.5×10^{-7}; the midpoint rule used before July 2026 erred by 6–9% at on-resonance cells (round 7, commits `53d74f2`–`ca816a4`).

Coupling cache.—Within a fit at fixed m_{Z_D} the rate grid depends on the couplings only through the total coupling s, as $U(s) \approx I_A + I_B/s$ to $\sim 7 \times 10^{-8}$ over brackets of up to two decades. The cache fits this form from two exact grids and uses it only after checking it against a third (tolerance 3×10^{-7}). Until 2026-07-29 its minimum bracket width (0.3 decades) exceeded the coupling steps of the minimizer, so it never fired in real scans. We replaced that width with a numerical floor. The cache then served 83% of the evaluations of minimizer-like sequences at three mediator masses, with a worst error of 6.5×10^{-9} against fresh exact grids (commit `5e7877c`).

3.3 Cosmological mixing

The eigenvectors of $M/(2E(1+z)) + A(z)S$ depend on E and z only through $\sigma = 2E(1+z)A(z)$, because eigenvectors are unchanged by an overall factor. Therefore, we tabulate the mixing probabilities on 4096 nodes in $\log \sigma$ and interpolate, instead of computing $\sim 10^6$ eigenvector sets per run. Our table agrees with direct computation to 2.7×10^{-14} at a stress point with a level crossing in range (`README.md`, Sec. 12).

3.4 Coherent propagation

The evolution operator of each step, $\exp(-iH\Delta x)$ for a 3×3 Hermitian H, is evaluated in closed form (Newton divided differences on the traceless part, with a separate treatment of nearly degenerate eigenvalues; `src/dp_propagator.py`). When the potential is so large that double precision cannot follow the oscillation phase, the propagator switches to an exactly unitary projector form; the fluxes stay finite but carry only the decoherence-averaged information. This happens roughly where $m_{Z_D}^2 + m_D^2 \lesssim 3 \times 10^6 \, g_D \epsilon' \ \mathrm{eV}^2$, i.e., below $m_{Z_D} \approx 5$ eV for the default couplings ($g_D = 10^{-4}$, $\epsilon' = 0.1$); the analysis warns when its scan window reaches this region.

3.5 Earth steps

In each Earth step we use the exact average of the PREM density over the step, integrated layer by layer with the layer crossings in closed form. The accumulated phase then equals the integral of the potential along the chord for any number of steps. Fluxes computed with 8 and with 1000

steps agree to $\leq 4.5\times10^{-9}$, and their energy-averaged flavor ratios to $\leq 6.5\times10^{-10}$, at the three benchmark points of Sec. 4 that are not in the phase-noise regime (commit `e1b6660`). Since the scheme converges to the exact path-ordered result as the steps shrink, this independence of the step count shows that the result is converged. The earlier left-endpoint scheme accumulated a phase error of thousands of radians, which moved the fluxes by up to 10^{-2} whenever the step count changed. The default is 25 steps since 2026-07-29 (commit `95fef9f`, Table 4).

3.6 Parallelism and cost

Milky Way paths and Earth chords are independent; they run in a persistent pool of processes started with `spawn`, the start method the analysis requires. The cosmological phase and the thermal grids use threads (numba, TBB layer). On the development laptop (Intel i5-1334U, 8 cores), a solver run at the defaults takes 0.32 s (best of three). One likelihood evaluation at the analysis settings takes 0.39 s on one core, 0.29 s when the coupling cache applies, and 0.17 s with the spatial phases spread over 8 cores (medians, 2026-07-29). Before the step count fell from 100 to 25, the Earth phase took 66% of a single-core evaluation and the Milky Way phase 18%.

4 Validation

4.1 The four benchmark points

Every change that can affect the physics is checked against the full solver at production settings ($N_E = N_z = 1000$, 6 paths, 12 chords, 4 cores) at four parameter points, with the reference and the modified code in separate processes:

Point	m_{Z_D} [eV]	g_D	ϵ'
below	3×10^5	10^{-4}	0.1
inband	2×10^6	10^{-4}	0.1
strong	2×10^7	10^{-2}	0.1
cosmoX2	40	10^{-4}	0.1

At "below" all three absorption resonances lie under the energy range; at "inband" they lie inside it; "strong" has a hundred times the coupling. At "cosmoX2" the mediator is light, so the potential dominates the Earth transit and the accumulated phase is large enough that a change in the last floating-point digit moves the pointwise fluxes by $\sim 10^{-3}$.

Acceptance rules.—We compare two quantities: the largest relative flux difference over all energies and flavors, and the energy-averaged flavor ratios. A change that should not alter the arithmetic must reproduce "below", "inband" and "strong" to the floor at which two runs of identical code agree, ~ 2–7×10^{-12}. The "strong" point has an additional common-mode floor of $\sim 7.5\times10^{-8}$ under any reordering of the last floating-point digit, with its flavor ratios at $\lesssim 10^{-12}$. We judge "cosmoX2" on the energy-averaged flavor ratios and the energies of the absorption dips only. For the same reason, any recompilation of the spatial kernels changes the results at the last digit: two textually identical kernels compiled separately differed by 3.7×10^{-16}.

4.2 Record of validated changes

5 Statistical analysis: current state

5.1 Likelihood and parameters

The analysis (`FrequentistAnalysis/`) predicts the flavor composition at Earth, $f_\alpha = \int \Phi_\alpha\, dE / \int \Phi\, dE$ over 10 TeV–30 PeV, and compares it with a projected measurement:

Table 4: **Changes checked at the benchmark points.** Differences are the largest relative flux difference and the largest change of an energy-averaged flavor fraction.

Date, commit	Change	Result
2026-07-29 `e1b6660`	Phase-exact Earth stepping	Step-count independent to 4.5×10^{-9} (Sec. 3.5)
2026-07-29 `0d6392c`	Solver split into modules	Identical at inband, strong, cosmoX2; below at the 7×10^{-12} floor
2026-07-29 `95fef9f`	Earth steps $100 \to 25$	below 1.0×10^{-9} / 1.2×10^{-10}; inband 2.1×10^{-11} / 2.3×10^{-12}; strong 6.2×10^{-12} / 8.3×10^{-13}; cosmoX2 ratios 2.9×10^{-3}, dips unchanged
2026-10-12 `5ecf10e`	Relic temperature corrected	A physics change; effects in Appendix A.1
2026-10-12 `0db8111`	Cosmological and Galactic neutron densities	No change at the four points beyond their floors; effects at light mediators in Appendix A.2
2026-10-12 `9c025e6`	Neutron fractions along Earth chords	Flavor fractions move by up to 0.026, as much as adding one chord (Appendix A.3)
2026-10-12 (this revision)	Point sources and cold disc gas	No change at the four points beyond their floors; effects at light mediators in Appendix A.4

an asymmetric Gaussian fitted to the expected 2050 sensitivity of IceCube with IceCube-Gen2, KM3NeT, Baikal-GVD, P-ONE and HUNT, centred on full pion decay ("case 5" in `core/params_detector_flavor_likelihood.py`). Detector uncertainties enter only through the width of that projection. In the shipped configurations (`config/pion_2050_case_5_fixed_nuisance*.json`) four parameters float, m_{Z_D}, g_D, u_e and u_μ; everything else is fixed, including the mixing parameters, the source composition, the spectral index and the lightest neutrino mass.

Limitations we found.—The flavor fractions are plain integrals of the flux; a detector would weight them by its effective area and cross sections, which rise with energy. For an $E^{-2.5}$ spectrum the plain integral is dominated by energies near 10 TeV. The event-rate calculation of Sec. 6.1 removes this approximation. The configurations also define Gaussian priors for the mixing parameters, which are inactive because those parameters are fixed; they would not work if released (Appendix A.5).

5.2 Scans and limits

One-dimensional profiles fix the parameter of interest on a grid and minimize over the others with Minuit. Each grid point starts from its neighbor's solution: in a 100-point scan of m_{Z_D} (2026-07-29, 8 processes) these points needed a median of 26 evaluations, against 37 for the first point of each segment, which starts from the global fit. Repair passes re-minimize points that stand out from their neighbors; on that scan they found none. Exclusion planes (`FrequentistAnalysis/src/run_limits.py`) fix the pair (m_{Z_D}, coupling) on a 20×20 grid and read the limits off the profiled $\Delta\chi^2$ at 2.71 and 3.84, the two-sided 90% and 95% values for one parameter. Scans and planes save each completed segment, so an interrupted run resumes; the initial global fit of the profile scans is saved as well.

What we have not shown.—Nothing yet checks that the intervals have their stated coverage. Near the boundaries of the parameter space (couplings at zero, the lightest mass at zero) the asymptotic thresholds can fail. For an upper limit on a coupling bounded at zero the one-sided values are 1.64 and 2.71, so our current limits are probably conservative. Section 6.3 describes the tests.

6 Planned work

6.1 Event rates at IceCube

We will compute the event distributions in reconstructed energy for two event classes, from public IceCube responses, in a new package (`IceCubeRates/`).

Northern-Hemisphere muons.—Through-going muon tracks from below the horizon, where the Earth removes the atmospheric muon background. The response is IceCube's IceTracks-DR2 release (2008–2022, doi:10.7910/DVN/MMIIZA), which provides per-season muon-neutrino effective areas and smearing tables. For each bin of true energy and declination, a smearing table gives the fraction of events in each bin of reconstructed muon energy, angular distance from the true direction (point-spread function) and estimated angular error. Its 41 declination bins replace the 3 of the 2018 release, which matters near the horizon. We fold the flux with the PLEnuM framework (github.com/PLEnuM-group/Plenum), which also places detectors at other sites; our adapter converts the DR2 tables into PLEnuM's grids.

Starting events.—Cascades, starting tracks and double cascades, from IceCube's HESE 7.5-year release (github.com/icecube/HESE-7-year-data-release). It is the only public simulation of individual starting events that resolves all three morphologies. Each simulated event carries its flavor (neutrino or antineutrino), its interaction (charged current, neutral current or Glashow resonance), its reconstructed deposited energy and its morphology. It also carries a weight per unit flux, together with the atmospheric fluxes and self-veto probabilities at its energy and direction. We reweight each simulated event by our flux. For IceCube-Gen2 we scale these weights by the ratio of the Gen2 and IceCube effective areas published for the Gen2 design (doi:10.7910/DVN/XANWAT), per morphology, flavor, energy and zenith angle. Those tables carry no energy resolution, so the IceCube response supplies it.

Earth absorption and direction.—Both responses already contain absorption in the Earth, computed by IceCube with its nominal cross sections. We checked this in the DR2 tables: at 1 PeV the effective area for straight up-going neutrinos is 20 times smaller than at the horizon, and at 10 PeV 120 times smaller, while at 10 TeV it is nearly uniform. Our solver computes the other effect of the Earth, the flavor change caused by the dark-portal potential, which involves no cross section. Because collisions remove all flavors nearly equally, the two act as independent factors along each path; the Glashow resonance, ν_τ regeneration and neutral-current scatterings that continue at lower energy break this slightly. Since both factors depend on direction, the solver must return the flux per Earth chord, and must add down-going directions, instead of the average it returns now.

6.2 Systematic uncertainties

The fits will vary the following, agreed on 2026-10-12. *Astrophysical flux:* its normalization and spectral index. *Source composition:* $f_{e,S}$, with $f_{\tau,S} = 0$. *Atmospheric fluxes:* the conventional normalization, a tilt of the cosmic-ray spectrum and the twelve hadron-production parameters of Barr et al. above 30 GeV (pion regions G, H, I and kaon regions W, Y, Z, each for both charges). We compute their effects with MCEq (SIBYLL-2.3d and the H3a cosmic-ray model), exactly as in the NuFlavorSingleSource project. The prompt and atmospheric-muon normalizations and the self-veto vary as well. *Mixing parameters:* all six, constrained by the joint $\Delta\chi^2$ of NuFIT 6.1 with atmospheric data, per ordering. We use the three-dimensional table over $(\sin^2\theta_{23}, \Delta m^2_{31}, \delta_{\mathrm{CP}})$, the two-dimensional table over $(\sin^2\theta_{12}, \Delta m^2_{21})$ and the one-dimensional table over $\sin^2\theta_{13}$, so that each parameter appears in exactly one table. *Lightest neutrino mass:* free, with a prior on the sum of the masses.

The prior on the mass sum.—As in the GeoneutrinoDecay project, our prior is a half-Gaussian on the sum, which we derive from the lightest mass and the splittings. It adds $(\Sigma m_\nu/\sigma)^2$ to the χ^2, with $\sigma = L/\Phi^{-1}(0.95)$ for a published 95% upper limit L. We use

$L = 0.055$ eV, from the cosmic microwave background (Planck, ACT and SPT-3G) with the baryon acoustic oscillations of DESI DR2 in ΛCDM, with a prior on Σm_ν that starts at zero (SPT-3G D1, arXiv:2506.20707, Eq. 96). Then $\sigma = 0.0334$ eV. The smallest sum allowed in normal ordering, 0.0587 eV, costs $\Delta\chi^2 = 3.08$; in inverted ordering, 0.0995 eV costs 8.85. Both orderings are fitted. As a check, this prior excludes the two minimal sums at 96.0% and 99.9%, against 96.6% and 99.9% stated in the paper.

Not varied.—The cross sections are fixed by the public responses; varying them would require generating our own effective areas. The detector-systematic corrections of the HESE release (optical efficiency, hole ice, ice anisotropy) are deferred, because they need the PHOTOSPLINE package.

6.3 Statistical soundness

We will establish that the analysis is sound in the order used in the GeoneutrinoDecay project. *Closure.*—Each response must reproduce a published IceCube fit before it is used for new physics; the HESE release includes its own power-law fit. *Injection and recovery.*—On Asimov data with known dark-portal parameters, and without a signal, the fit must recover the truth without bias. *Fits.*—Then the fits with the current profile-likelihood procedure. *Coverage.*—We generate pseudo-experiments with Poisson-fluctuated counts and fit each exactly like the data, 200 per point (which gives ±2% on a 10% rate). The points are the no-signal hypothesis, where we compare the test statistic with its asymptotic distribution, and the expected limit and half a decade beyond it. We add the parameter boundaries, the region where the mediator mass and the lightest mass trade off in setting the resonance energy, and the low-count corners (double cascades, highest-energy tracks). The pseudo-experiments change only the data, so the solver predictions on the scan grid are computed once and reused. *Decision.*—If no point under-covers, we keep the profile likelihood; otherwise we construct Feldman–Cousins intervals with PyFC (github.com/mbustama/FeldmanCousins), whose likelihood also accounts for the finite size of the HESE simulation.

The projected flavor analysis.—It has no counts to fluctuate; its pseudo-experiments must shift the centre of the projected measurement by its own width, which is an approximation we will state with its results.

6.4 Computation

Our heavy runs go to Tycho, the cluster of the Niels Bohr Institute, set up as in the GRBsim and NuFlavorSingleSource projects. The repository and a conda environment live on the shared file system, and the job scripts read their core count from SLURM. Jobs take 16 cores by default and up to 32. Before the first run, the solver must size its process pools from the cores the job was given; it now uses the core count of the whole node, so a 16-core job would start 127 processes. The first job will repeat the benchmark validation of Sec. 4 against the laptop's results.

7 Open items

Earth-chord sampling floor.—In the Earth the dark-portal potential exceeds the vacuum term by $\sim 10^9$ at MeV mediator masses, so the phase accumulated along a chord is $\sim 10^8$ rad. These phases are effectively random from chord to chord, and the residual ν_e–ν_μ conversion left by a finite chord average is sampling noise. We measured it with the code of commit `0db8111`: adding a thirteenth chord to the default twelve moves the energy-averaged flavor fractions by 0.048 (below), 0.022 (inband), 0.007 (strong) and 1.6×10^{-3} (cosmoX2). Scaled to the analysis's 50 chords this is $\sim 10^{-2}$, which exceeds the projected 2050 width of f_μ (≈ 0.005). Over a continuum of zenith angles the physical answer is the decoherent average, which a finite

chord sum only approximates; an analytic average over the phases, where they vary by many cycles across a zenith bin, would remove this floor. **[Open, 2026-10-12]**

Implementation queue.—In order: the core-count fix, the Tycho setup and its validation job, the per-direction solver output, the muon channel, the starting-event channel with the systematic uncertainties, then the statistical tests of Sec. 6.3. The details are in `docs/dev/WORK_PLAN.md`.

A Corrections

A.1 The relic-neutrino temperature (2026-10-12)

The default $T_{\nu,0}$ was 1.95×10^{-4} eV, which is 2.26 K: the value "1.95 K" had been entered as if it were in eV. The correct value is 1.676389×10^{-4} eV, so the relic neutrinos were 16% too hot (commit `5ecf10e`). We measured the effect on the absorption rates directly. Above every resonance, where the relics act as if at rest, the old temperature raised the rates by a factor of 1.553–1.570 across the energy range at the "below" point, close to $(1.95/1.676389)^3 = 1.574$. Across a resonant dip it raised the energy-integrated absorption by 1.573, the depth of the dip by 1.35–1.38 (against $(1.95/1.676389)^2 = 1.353$) and its width by 5–16%. Over the energies and redshifts of the "inband" grid, where the shapes of the dips also depend on the temperature, the ratio ranged from 1.33 to 5.4.

Effect on the fluxes.—At the benchmark points the fluxes changed by up to a factor of ~ 130 at individual energies. At "below" and "cosmoX2", where the flux is strongly absorbed across the energy range, the median change was a factor of ~ 30, because a 57% change of the optical depth τ changes $e^{-\tau}$ by large factors once τ exceeds a few. Absorption features above the resonances moved down in energy, by 0.195 decades at "below" and "cosmoX2" (the relic-density factor, because the optical depth there scales as density over energy) and by ~ 0.15 decades at "inband". The resonant dips stayed at their energies (E_{res} does not depend on the temperature) but became shallower: the flux at the deepest "inband" dip rose by a factor of 3.6, at the deepest "strong" dip by 36%. The energy-averaged flavor fractions moved by at most 3.8×10^{-4} (below), 1.8×10^{-4} (inband), 7×10^{-9} (strong) and 6.7×10^{-5} (cosmoX2). Therefore, results of the flavor-ratio analysis obtained before the correction are almost unchanged, while spectra and event rates change substantially.

A.2 The cosmological and Galactic neutron densities (2026-10-12)

Cosmological.—The default $n_{n,0}$ was 2.5×10^{-7} cm^{-3}, documented as the neutron density but equal to the total baryon density (2.515×10^{-7} cm^{-3}, PDG 2024). The potential (T9) is sourced by neutrons alone: the vector couplings of the Z to protons ($\frac{1}{2} - 2\sin^2\theta_W$) and to electrons ($-\frac{1}{2} + 2\sin^2\theta_W$) cancel in neutral matter. The mediator Z_D inherits them through the mass mixing. Neutrons sit almost entirely in helium-4, which holds a fraction $Y_p = 0.2458 \pm 0.0013$ of the baryons (LBT Y_p project, 2026), half of them neutrons; deuterium adds $\sim 2 \times 10^{-5}$ per baryon. Therefore, $n_{n,0} = n_b Y_p/2 = 3.091 \times 10^{-8}$ cm^{-3}, and the cosmological potential was 8.1 times too strong.

Galactic.—The Galactic model converted the stellar mass density at half a neutron per nucleon (the factor $20.235 = 0.5 \times 40.464$ nucleons cm^{-3} per $M_\odot$ pc^{-3}) and the hot-halo density of Miller and Bregman (2013), which is an electron density, at 0.5 neutrons per electron. We now use 0.20 neutrons per nucleon for stars. Locally, stars and remnants amount to $33.4\,M_\odot$ pc^{-2}, of which 81% are visible stars, 15% white dwarfs and 4% brown dwarfs (McKee, Parravano and Hollenbach 2015). With $(Y + Z)/2 = 0.146$ for disc stars ($Y \approx 0.275$, $Z \approx 0.017$), 0.5 for carbon–oxygen white dwarfs and 0.142 for brown dwarfs, the weighted fraction is 0.20. For the hot gas ($Y \approx 0.26$, $Z \approx 0.3\,Z_\odot$, fully ionized) we use $(Y + Z)/2$ neutrons per $1 - (Y + Z)/2$

electrons, i.e., 0.15. The Galactic potential was thus 2.5 times (stars) and 3.3 times (gas) too strong.

Effect.—At the four benchmark points neither change alters the results beyond their floors: "below", "inband" and "strong" agree to $\leq 4 \times 10^{-11}$ in flux and $\leq 10^{-13}$ in flavor fractions; "cosmoX2" moves by 1.4×10^{-3} pointwise and 1.0×10^{-4} in the energy-averaged fractions, its known sensitivity to the last digit. At mediator masses of order MeV the coherent potentials are $\sim 10^{-22}$ of the vacuum term, so only the resonant absorption acts. The corrections matter for light mediators, where the potential grows as $1/m_{Z_D}^2$; we compared the old and new code at three such masses, with the default couplings. At $m_{Z_D} = 10^{-2}$ eV the total flux changes by up to 32%, the flavor fractions at single energies by up to 0.074 and the energy-averaged fractions by up to 0.013 (f_τ: 0.315 $\to$ 0.328). At 10^{-5} eV the changes are 69%, 0.144 and 0.004. At 10^{-8} eV, where the potential is so large that the flavor evolution is fully averaged, they are 3%, 0.008 and 10^{-4}.

A.3 Neutron fractions along Earth chords (2026-10-12)

Along Earth chords the code counted half of the nucleons as neutrons in every PREM layer (3.011×10^{23} g^{-1}). For neutral matter the neutrons per nucleon are $1 - Y_e$, where Y_e is the electron (equal to proton) fraction. We now use $Y_e = 0.4656$ in the iron–nickel core and 0.4957 in the silicate mantle and crust, the values used with PREM in neutrino-oscillation tomography, and the composition of water (8/18.015 neutrons per molecular mass) for PREM's top layer. The core was thus 6.9% low, the mantle 0.9% low and the water layer 12.6% high. The regional boundaries coincide with PREM layer boundaries, so the phase-exact Earth steps stay exact (test `test_earth_neutrons_per_gram_by_layer`).

Effect.—The four benchmark points move: the energy-averaged ν_e and ν_μ fractions change by 0.025 (below), 0.026 (inband), 0.009 (strong) and 0.032 (cosmoX2), with the ν_τ fraction unchanged, and the flux at single energies by up to 21%. A change of a few per cent in the potential shifts the chord phases by $\sim 10^6$ rad, which redraws the residual conversion left by the 12-chord average. Adding one chord to the old code moves the same fractions by comparable amounts (Sec. 7). At light mediators the changes are smaller: up to 3.8×10^{-3} in the energy-averaged fractions at $m_{Z_D} = 10^{-2}$ eV and below 10^{-5} at 10^{-5} and 10^{-8} eV.

A.4 Point sources and cold disc gas (2026-10-12)

The Sun, the Moon and the Earth enter the Milky Way and Earth phases as point sources of a Yukawa potential, which reaches the detector only when the mediator's range, $1/\sqrt{m_{Z_D}^2 + m_D^2}$, spans the distance to the body. Their neutron numbers were inconsistent: 0.248 neutrons per nucleon for both the Sun and the Moon, 0.496 for the Earth. The Sun's present bulk composition is the protosolar one (X, Y, $Z = 0.7154$, 0.2703, 0.0142; Asplund et al. 2009) plus the $\sim 0.038\,M_\odot$ of hydrogen it has burned into helium, which we estimate from its luminosity integrated over 4.57 Gyr; that gives $Y = 0.308$ and $(Y + Z)/2 = 0.161$ neutrons per nucleon, 1.929×10^{56} in all. The Moon is silicate rock with a small iron core, 0.505 per nucleon, 2.232×10^{49}. For the Earth we integrate PREM with the per-layer fractions of the chords (Appendix A.3); PREM's mass comes out at 5.973×10^{27} g, within 0.02% of the Earth's, with 1.849×10^{51} neutrons (0.514 per nucleon).

The Galactic model had no cold gas. We add the HI and H_2 discs of McMillan (2017, arXiv:1608.00971), $\rho = \Sigma_0/(4z_d)\, e^{-R_m/R - R/R_d}\, \mathrm{sech}^2(z/2z_d)$ with $(\Sigma_0, R_d, R_m, z_d) = (53.1\,M_\odot\,\mathrm{pc}^{-2}, 7, 4, 0.085$ kpc) and $(2180\,M_\odot\,\mathrm{pc}^{-2}, 1.5, 12, 0.045$ kpc). Their surface densities already include helium and metals; at the Sun they give 10 and $2\,M_\odot$ pc^{-2}, against the $13.7\,M_\odot$ pc^{-2} that McKee et al. (2015) measure in all gas. We convert them at $(Y+Z)/2 = 0.142$ neutrons per nucleon (protosolar composition).

Effect.—At the four benchmark points nothing changes beyond the floors ($\leq 4 \times 10^{-11}$ in flux; "cosmoX2" 9×10^{-4} pointwise), because at MeV masses the point sources are out of range and the Galactic potential is negligible. At light mediators the changes appear at individual energies: at $m_{Z_D} = 10^{-2}$ eV up to 18% in flux and 0.064 in the flavor fractions, at 10^{-8} eV up to 3% and 0.008. We also ran $m_{Z_D} = 10^{-20}$ eV, where the bodies are within range: with Debye screening off the flux changes by up to 27% and the flavor fractions by up to 0.13; with $g_D = 10^{-12}$, by 1.6% and 0.004. The energy-averaged fractions move by at most 1.2×10^{-3} in all these runs.

A.5 The mixing-parameter priors (found 2026-10-12, not yet fixed)

The Gaussian priors in the `fallbacks` block of the analysis configurations have never been applied, for three reasons. First, the configurations name the shape `"gaussian"`, while the code accepts only `"gauss"`. Second, the entries `dcp` and `dm3l_sq` do not match the parameter names `d_cp_deg` and `dm31_sq`. Third, the prior centres disagree with the fixed values (δ_{CP} 212° against 132°, Δm^2_{21} 7.537×10^{-5} against 7.41×10^{-5} eV^2, which is 3.4σ of the prior). In addition, the branch for angles in degrees converts a value that is already in degrees. The NuFIT 6.1 constraints of Sec. 6.2 will replace these priors.

A.6 Earlier corrections

Thermal kernel (July 2026).—The midpoint-rule momentum integral erred by 6–9% at on-resonance cells; validation against the previous code could not reveal this, since it measures only differences. A dense reference integration did. The semi-analytic kernel replaced it, which changed the fluxes at the dips by up to 15% with flavor ratios within 10^{-6} and dip energies unchanged (round 7).

Earth steps (July 2026).—The left-endpoint stepping scattered the results at the 10^{-2} level under any change of the step count (Sec. 3.5); in-band energy-averaged flavor ratios carried a 1.7×10^{-2} ambiguity at the analysis settings. The phase-exact scheme removed it.

Coupling cache (July 2026).—The cache never activated in real scans (Sec. 3.2); its narrower brackets fixed that without loss of accuracy.